SPI frame 1: TX = 0x9280 on MOSI, RX = 0x91F0 on MISO (16 bits, MSB first)

SPI_Master, CPOL=1 / CPHA=0. Bit k = k-th SCLK falling edge: the ADC reads MOSI and the FPGA reads MISO.

After bit	MOSI sent	tx_shift_v (what's left to send)	MISO received	rx_shift_v (what's arrived)
start	-	1001 0010 1000 0000 (9280)	-	0000 0000 0000 0000 (0)
1	1	0010 0101 0000 0000 (2500)	1	0000 0000 0000 0001 (1)
2	0	0100 1010 0000 0000 (4A00)	0	0000 0000 0000 0010 (2)
3	0	1001 0100 0000 0000 (9400)	0	0000 0000 0000 0100 (4)
4	1	0010 1000 0000 0000 (2800)	1	0000 0000 0000 1001 (9)
5	0	0101 0000 0000 0000 (5000)	0	0000 0000 0001 0010 (12)
6	0	1010 0000 0000 0000 (A000)	0	0000 0000 0010 0100 (24)
7	1	0100 0000 0000 0000 (4000)	0	0000 0000 0100 1000 (48)
8	0	1000 0000 0000 0000 (8000)	1	0000 0000 1001 0001 (91)
9	1	0000 0000 0000 0000 (0)	1	0000 0001 0010 0011 (123)
10	0	0000 0000 0000 0000 (0)	1	0000 0010 0100 0111 (247)
11	0	0000 0000 0000 0000 (0)	1	0000 0100 1000 1111 (48F)
12	0	0000 0000 0000 0000 (0)	1	0000 1001 0001 1111 (91F)
13	0	0000 0000 0000 0000 (0)	0	0001 0010 0011 1110 (123E)
14	0	0000 0000 0000 0000 (0)	0	0010 0100 0111 1100 (247C)
15	0	0000 0000 0000 0000 (0)	0	0100 1000 1111 1000 (48F8)
16	0	0000 0000 0000 0000 (0)	0	**1001 0001 1111 0000 (91F0)**

How to read it:

- MOSI sent (top to bottom) = 1001 0010 1000 0000 = 0x9280, the command. MISO received (top to bottom) = 1001 0001 1111 0000 = 0x91F0, the result.
- Each tx_shift_v row = previous row moved 1 place left, '0' added at the right; the bit that falls off the left is that row's 'MOSI sent'.
- Each rx_shift_v row = previous row moved 1 place left, that row's 'MISO received' added at the right.
- TX empties from the top (bit 15 goes out first), RX fills from the bottom; after bit 16 the received word is right-justified in bits 15:0.
- Bits 1-6: MOSI and MISO carry the same bits (both words start with 1001 00); they differ from bit 7 on.